CREDITCOIN BUIDL HACKATHON

CredLock

Cross-chain collateral safety for Creditcoin financing

An asset can be pledged as collateral on one chain, then presented to Creditcoin for financing. The second lender cannot see the first pledge.

CredLock checks the collateral state before Creditcoin lends. Clear means go. Pledged means the financing transaction itself refuses.

PLEDGED ELSEWHERE

Asset pledged on Sepolia

v

Attestcoin proves the fact

v

Creditcoin verifies the proof inside the Borrow transaction

v

BLOCKED - AssetEncumbered

CLEAR

Asset clear on Sepolia

v

Attestcoin proves the fact

v

Creditcoin verifies the proof inside the Borrow transaction

v

ALLOWED - Financing executes

HOW IT WORKS

Borrow carries the proof.

The proof is not displayed by the frontend and trusted. The Borrow transaction carries a fresh Attestcoin proof to the Creditcoin gate. The gate verifies it inline through the BlockProver precompile and derives the decision from the proven transaction itself.

ARCHITECTURE

Sepolia SourcePledgeRegistry - emits AssetRegistered / AssetPledged

v

Collateral fact for one asset id

v

Attestcoin - Merkle inclusion + continuity proof (usc-sdk)

v

Creditcoin CredLockGate.requestFinancingWithProof

v

CLEAR -> FINANCE | PLEDGED -> BLOCK

DEPLOYED CONFIGURATION

```
Gate (Creditcoin 102031): 0xF1028099d9CeE27b355159beCD111c8f9A409F67
Registry (Sepolia):       0xae543bb778df66774585b99d9135676bc5d99c2a
Source chainKey: 1  |  Verifier precompile: 0x0000..0FD2
```

BUILT WITH

Solidity + Foundry (20 contract tests, all green). @gluwa/usc-sdk proof builder. EvmV1Decoder event checks. Next.js + wagmi UI where every write is signed by the user wallet. No server key, no mocks on the verification path.

PROOF IT WORKS

Two assets, two outcomes.

SCENARIO A - CLEAR ASSET BORROWS

Asset 0x6349...0ae9e3. Registered on Sepolia, never pledged. Borrow submitted the fresh registration proof inside the financing transaction. Verified on-chain, financing executed, verdict ALLOW, financed true.

```
Sepolia register: 0x1f3fee534c5f757ebd70b979102d43ef01d6baea07a921ff59db72b3c6c1c6d4
Creditcoin borrow: 0xab138a65bc019554e1bf5a161451ece5b977791df90c0e2610e8f7188fd16c7d (block
5470652)
```

SCENARIO B - PLEDGED ASSET REVERTS

Asset 0x8bfc...024d9 (uyscutty). Registered, then pledged on Sepolia. Borrow submitted the fresh pledge proof. The gate reverted with AssetEncumbered (selector 0x01d7f74f) on a live node call, and the forge suite covers the mined revert path. No financing possible through this gate.

```
Sepolia pledge:   0xd0625d429caab95496629bde3c50c865ac9d4c04bc2e2881d7e39fcd2f23c107 (block
11684010)
Revert reason:    AssetEncumbered(), proven live + 20/20 forge tests
```

LINKS

Repository https://github.com/uyscuttyy/CredLock

Gate contract (Blockscout) https://creditcoin-testnet.blockscout.com/address/0xF1028099d9CeE27b355159beCD111c8f9A409F67

Registry (Etherscan) https://sepolia.etherscan.io/address/0xae543bb778df66774585b99d9135676bc5d99c2a

Borrow tx, scenario A https://creditcoin-testnet.blockscout.com/tx/0xab138a65bc019554e1bf5a161451ece5b977791df90c0e2610e8f7188fd16c7d

Register tx, scenario A https://sepolia.etherscan.io/tx/0x1f3fee534c5f757ebd70b979102d43ef01d6baea07a921ff59db72b3c6c1c6d4

Pledge tx, scenario B https://sepolia.etherscan.io/tx/0xd0625d429caab95496629bde3c50c865ac9d4c04bc2e2881d7e39fcd2f23c107

Proof builder https://prover.cc3-testnet.creditcoin.network

Live app runs locally from the repository (see README). No staging URL is claimed.